Unit 6 Pre-Test – Lines and Planes Unit 6

MCV4U

Jensen

1) Write a vector equation for the line given its direction vector $\vec{m} = [3,1]$ and point on the line $P(2,7)$.

$$[x,y] = [2,7] + t[3,1]$$

2) Write the vector equation of the line through each pair of points.

a) $A(1,7)$ and $B(4,10)$

$\vec{r}_0 = [1,7]$

$\vec{AB} = [3,3] = 3[1,1]$

$[x,y] = [1,7] + t[1,1]$

b) $A(-3,5)$ and $B(-2,-8)$

$\vec{r}_0 = [-3,5]$

$\vec{m} = \vec{AB} = [1,-13]$

$[x,y] = [-3,5] + t[1,-13]$

3) Determine if the point $P(\overset{x}{-1},\overset{y}{11})$ is on the line $[x,y] = [3,1] + t[-2,5]$

$$l: \begin{cases} x = 3-2t \\ y = 1+5t \end{cases}$$

$-1 = 3-2t$ $\quad 11 = 1+5t$

$-4 = -2t$ $\quad 10 = 5t$

$t = 2$ $\quad t = 2$

$\therefore P(-1,11)$ is on the line.

4) Write the parametric equation of each vector equation

a) $[x, y] = [10,6] + t[13,1]$

$l: \begin{cases} x = 10 + 13t \\ y = 6 + t \end{cases}$

b) $[x, y, z] = [3,0,-1] + t[6,-9,1]$

$l: \begin{cases} x = 3 + 6t \\ y = -9t \\ z = -1 + t \end{cases}$

5) Write the vector equation of each parametric equation

a) $\ell: \begin{cases} x = 1 + 4t \\ y = -6 + t \\ z = 2 - 2t \end{cases}$

$[x,y,z] = [1,-6,2] + t[4,1,-2]$

b) $\ell: \begin{cases} x = 7 \\ y = -t \\ z = 0 \end{cases}$

$[x,y,z] = [7,0,0] + t[0,-1,0]$

6) Given the points $A(3,4,-5)$ and $B(9,-2,7)$, write a vector equation and the parametric equations of the line through A and B.

$\vec{r}_0 = [3,4,-5]$

$\vec{m} = \vec{AB} = [6,-6,12] = 6[1,-1,2]$

vector:

$[x,y,z] = [3,4,-5] + t[1,-1,2]$

parametric:

$l: \begin{cases} x = 3 + t \\ y = 4 - t \\ z = -5 + 2t \end{cases}$

7) Determine if $\ell_1: \begin{cases} x = 3 - 2t \\ y = 4 + t \end{cases}$ and $\ell_2: 2x + 4y - 7 = 0$ are parallel, perpendicular, or neither. Explain.

$\vec{n}_1 = [-2,1]$

$\vec{n}_2 = [2,4]$

$\vec{n}_1 \cdot \vec{n}_2 = [-2,1] \cdot [2,4]$

$= -2(2) + 1(4)$

$= 0$

$\vec{n}_2$ is perpendicular to both l_1 and l_2; therefore l_1 and l_2 are parallel.

8) Determine a vector equation of each line.

a) perpendicular to line $4x - 3y = 17$ and through point $P(-2,4)$

$$[x, y] = [-2,4] + t[4,-3]$$

b) parallel to the z-axis and through point $P(1,5,10)$

$$[x,y,z] = [1,5,10] + t[0,0,1]$$

c) parallel to $[x, y, z] = [3,3,0] + t[3, -5, -9]$ with an x-intercept of -10

$$[x,y,z] = [-10,0,0] + t[3,-5,-9]$$

d) with the same x-intercept as $[x, y, z] = [3,0,0] + t[4, -4,1]$ and the same z-intercept as $[x, y, z] = [6, -2, -3] + t[3, -1, -2]$

$$l_2: \begin{cases} x = 6+3t \\ y = -2-t \\ z = -3-2t \end{cases} \qquad \begin{array}{l} 0 = 6+3t \\ t = -2 \end{array} \qquad \begin{array}{l} 0 = -2-t \\ t = -2 \end{array} \qquad \begin{array}{l} z = -3-2(-2) \\ z = 1 \end{array}$$

x-int: $P_1(3,0,0)$

z-int: $P_2(0,0,1)$

$\vec{n} = \vec{P_2P_1} = [3,0,-1]$

$$[x,y,z] = [3,0,0] + t[3,0,-1]$$

9) Does the point $P(1,2,0)$ lie on the plane $4x + 3y - 5z = 10$?

LS

$= 4x + 3y - 5z$

$= 4(1) + 3(2) - 5(0)$

$= 10$

RS

$= 10$

$\therefore P(1,2,0)$ is on the line.

LS = RS

10) Find the x, y, and z-intercepts of each plane

a) $4x + 2y - 7z + 14 = 0$

x-int

$4x + 2(0) - 7(0) + 14 = 0$

$4x = -14$

$x = -\frac{7}{2}$

$(-\frac{7}{2}, 0, 0)$

y-int

$4(0) + 2y - 7(0) + 14 = 0$

$2y = -14$

$y = -7$

$(0, -7, 0)$

z-int

$4(0) + 2(0) - 7z + 14 = 0$

$14 = 7z$

$z = 2$

$(0, 0, 2)$

b) $3x + 6z + 18 = 0$

x-int

$3x + 6(0) + 18 = 0$

$3x = -18$

$x = -6$

$(-6, 0, 0)$

y-int

none

z-int

$3(0) + 6z + 18 = 0$

$6z = -18$

$z = -3$

$(0, 0, -3)$

11) Write the parametric equation of the plane $[x, y, z] = [0, -4, 1] + s[1, 10, -1] + t[0, 3, 4]$

$$\pi: \begin{cases} x = s \\ y = -4 + 10s + 3t \\ z = 1 - s + 4t \end{cases}$$

12) Write the vector equation of the plane π: $\begin{cases} x = 2 + s + 7t \\ y = 12s - 8t \\ z = 11 + 6s \end{cases}$

$$[x, y, z] = [2, 0, 11] + s[1, 12, 6] + t[7, -8, 0]$$

13) Determine if the point $P(-4,-13,10)$ is on the plane $[x,y,z] = [6,-7,10] + s[1,3,-1] + t[2,-2,1]$

Scalar Eq^n of plane:

$\vec{n} = \vec{m}_1 \times \vec{m}_2 = [\overset{A}{1}, \overset{B}{-3}, \overset{C}{-8}]$

3×-2
-1×1
1×2
3×-2

$Ax+By+Cz+D=0$
$1(6)-3(-7)-8(10)+D=0$
$D=53$
$x-3y-8z+53=0$

Check if $P(-4,-13,10)$ is on plane:

LS
$= x-3y-8z+53$
$= -4-3(-13)-8(10)+53$
$= 8$

RS
$= 0$

$LS \neq RS$

∴ $P(-4,-13,10)$ is NOT on plane.

14) Determine the x, y, and z-intercepts of $[x,y,z] = [1,8,6] + s[1,-12,-12] + t[2,4,-3]$

Scalar Eq^n:

$\vec{n} = \vec{m}_1 \times \vec{m}_2 = [84, -21, 28]$
$= 7[\overset{A}{12}, \overset{B}{-3}, \overset{C}{4}]$

-12×4
-12×-3
1×2
-12×4

$12x-3y+4z+D=0$
$12(1)-3(8)+4(6)+D=0$
$D=-12$

$12x-3y+4z-12=0$

x-int: $12x-3(0)+4(0)-12=0$
$12x=12$
$x=1$
$(1,0,0)$

y-int: $12(0)-3y+4(0)-12=0$
$-3y=12$
$y=-4$
$(0,-4,0)$

z-int: $12(0)-3(0)+4z-12=0$
$4z=12$
$z=3$
$(0,0,3)$

15) Write a vector and parametric equation of each plane

a) contains the points $A(1,3,-2)$, $B(3,-9,7)$, and $C(4,-4,5)$

$\vec{n}_1 = \vec{AB} = [2,-12,9]$

$\vec{n}_2 = \vec{BC} = [1,5,-2]$

$[x,y,z] = [1,3,-2] + s[2,-12,9] + t[1,5,-2]$

$$\pi: \begin{cases} x = 1+2s+t \\ y = 3-12s+5t \\ z = -2+9s-2t \end{cases}$$

b) has x-int 8, y-int -3, and z-int 2

$P_1(8,0,0)$, $P_2(0,-3,0)$, $P_3(0,0,2)$

$\vec{n}_1 = \vec{P_2P_1} = [8,3,0]$

$\vec{n}_2 = \vec{P_2P_3} = [0,3,2]$

$[x,y,z] = [8,0,0] + s[8,3,0] + t[0,3,2]$

$$\pi: \begin{cases} x = 8+8s \\ y = 3s+3t \\ z = 2t \end{cases}$$

16) Determine an equation for each plane

a) parallel to the xz-plane; and through the point $A(3,1,5)$

$y = 1$

b) containing the points $A(2,1,1)$, $B(5,-3,2)$, and $C(0,-1,4)$

$\vec{n}_1 = \vec{AB} = [3,-4,1]$

$\vec{n}_2 = \vec{CB} = [5,-2,-2]$

$[x,y,z] = [2,1,1] + s[3,-4,1] + t[5,-2,-2]$

17) Write the scalar equation of a plane given its normal $\vec{n} = [3,7,1]$ and a point $P(3,-6,4)$ on the plane.

$A\ B\ C \qquad x\ y\ z$

$Ax + By + Cz + D = 0$

$3(3) + 7(-6) + 1(4) + D = 0$

$D = 29$

$3x + 7y + z + 29 = 0$

18) Write a scalar equation of each plane, given its vector or parametric equation.

a) $[x, y, z] = [3,7,-5] + s[1,2,-1] + t[1,-2,3]$

$\vec{n} = \vec{m}_1 \times \vec{m}_2 = [4,-4,-4]$

$= 4[1,-1,-1]$ (A, B, C)

2 × −2 = 3; −1 × 3 = 1; 1 × 1 = −2; 2 × −2

$1x - 1y - 1z + D = 0$

$1(3) - 1(7) - 1(-5) + D = 0$

$D = -1$

$x - y - z - 1 = 0$

b) $\pi: \begin{cases} x = 3 - 2s + 2t \\ y = 1 + 3s + t \\ z = 5 - s - 2t \end{cases}$

$\vec{m}_1 = [-2,3,-1]$

$\vec{m}_2 = [2,1,-2]$

$\vec{r}_0 = [3,1,5]$

$\vec{n} = \vec{m}_1 \times \vec{m}_2 = [-5,-6,-8]$

$= -1[5,6,8]$

3 × 1 = −2; −1 × −2 = 2; −2 × 2 = 1; 3 × 1

$Ax + By + Cz + D = 0$

$5x + 6y + 8z + D = 0$

$5(3) + 6(1) + 8(5) + D = 0$

$D = -61$

$5x + 6y + 8z - 61 = 0$

19) Write a vector and scalar equation of each plane

a) perpendicular to the line $[x, y, z] = [2,4,-9] + t[3,5,-3]$ and including the point $P(4,-2,7)$

$\vec{n} = [3,5,-3]$

$\vec{m}_1 = [0,3,5]$

$\vec{m}_2 = [5,-3,0]$

vector: $[x,y,z] = [4,-2,7] + s[0,3,5] + t[5,-3,0]$

scalar:

$3x + 5y - 3z + D = 0$

$3(4) + 5(-2) - 3(7) + D = 0$

$D = 19$

$3x + 5y - 3z + 19 = 0$

b) parallel to the yz-plane and including the point $P(-1,-2,5)$

$\vec{n}_1 = [0,1,0]$

$\vec{n}_2 = [0,0,1]$

$\vec{n} = [1,0,0]$

vector: $[x,y,z] = [-1,-2,5] + s[0,1,0] + t[0,0,1]$

Scalar: $x + D = 0$

$-1 + D = 0$

$D = 1$

$x + 1 = 0$ OR $x = -1$

c) parallel to the plane $3x - 9y + z - 12 = 0$ and including the point $P(-3,7,1)$

$\vec{n} = [3,-9,1]$

$\vec{m}_1 = [0,1,9]$

$\vec{m}_2 = [1,0,-3]$

vector: $[x,y,z] = [-3,7,1] + s[0,1,9] + t[1,0,-3]$

Scalar: $3x - 9y + z + D = 0$

$3(-3) - 9(7) + 1 + D = 0$

$D = 71$

$3x - 9y + z + 71 = 0$

d) containing the lines $[x,y,z] = [-2,3,12] + s[-2,1,5]$ and $[x,y,z] = [1,-4,4] + t[-6,3,15]$

$P_1(-2,3,12)$

$P_2(1,-4,4)$

$\vec{m}_1 = \vec{P_1P_2} = [3,-7,-8]$

$\vec{m}_2 = [-2,1,5]$

Scalar: $\vec{n} = \vec{m}_1 \times \vec{m}_2 = [-27,1,-11]$

$-7 \quad 1$

$-8 \quad 5$

$3 \quad -2$

$-7 \quad 1$

$-27x + y - 11z + D = 0$

$-27(1) - 4 - 11(4) + D = 0$

$D = 75$

$-27x + y - 11z + 75 = 0$

$27x - y + 11z - 75 = 0$

vector:

$[x,y,z] = [-2,3,12] + s[3,-7,-8] + t[-2,1,5]$

20) Determine the angle between the planes $\pi_1: 3x + 2y + 5z = 5$ and $\pi_2: 4x - 3y - 2z = 2$

$\vec{n}_1 = [3, 2, 5]$

$\vec{n}_2 = [4, -3, -2]$

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\cos\theta = \frac{3(4) + 2(-3) + 5(-2)}{(\sqrt{38})(\sqrt{29})}$$

$$\cos\theta = \frac{-4}{(\sqrt{38})(\sqrt{29})}$$

$$\theta \approx 96.9^\circ$$

21) State the intersections of the following systems. Always try to predict the intersection before you spend time solving.

a) $\ell_1: [x, y] = [-12, -7] + s[8, -5]$
$\ell_2: [x, y] = [2, -1] + t[3, -2]$

$\ell_1: \begin{cases} x = -12 + 8s \\ y = -7 - 5s \end{cases}$

$\ell_2: \begin{cases} x = 2 + 3t \\ y = -1 - 2t \end{cases}$

$2 + 3t = -12 + 8s \qquad -1 - 2t = -7 - 5s$

① $14 = -3t + 8s$ ② $6 = 2t - 5s$

$2 \times$ ① $28 = -6t + 16s$
$3 \times$ ② $18 = 6t - 15s$
$46 = s$

POI:

$x = -12 + 8(46) = 356$

$y = -7 - 5(46) = -237$

$(356, -237)$

b) $\ell_1: [x, y, z] = [1, -1, 0] + s[2, -1, 3]$
$\ell_2: [x, y, z] = [1, 2, 3] + t[3, 2, 4]$

$\ell_1: \begin{cases} x = 1 + 2s \\ y = -1 - s \\ z = 3s \end{cases}$

$\ell_2: \begin{cases} x = 1 + 3t \\ y = 2 + 2t \\ z = 3 + 4t \end{cases}$

$1 + 2s = 1 + 3t \qquad -1 - s = 2 + 2t \qquad 3s = 3 + 4t$

① $0 = 3t - 2s$ ② $-3 = 2t + s$ ③ $-3 = 4t - 3s$

① $0 = 3t - 2s$
$2 \times$ ② $-6 = 4t + 2s$
$-6 = 7t$
$t = \frac{-6}{7}$

$0 = 3\left(\frac{-6}{7}\right) - 2s$
$\frac{18}{7} = -2s$
$s = \frac{-9}{7}$

check $t = \frac{-6}{7}$ and $s = \frac{-9}{7}$ in ③:

LS $= -3$

RS $= 4t - 3s$
$= 4\left(\frac{-6}{7}\right) - 3\left(\frac{-9}{7}\right)$
$= \frac{-24}{7} + \frac{27}{7}$
$= \frac{3}{7}$

LS ≠ RS

∴ No solutions. Lines are skew.

c) $\ell_1: [x,y,z] = [7,2,-6] + s[2,1,-3]$
$\ell_2: [x,y,z] = [3,9,13] + t[1,5,5]$

$7+2s=3+t \quad 2+s=9+5t \quad -6-3s=13+5t$

① $4 = t-2s$ ② $-7 = 5t-s$ ③ $-19 = 5t+3s$

$\ell_1: \begin{cases} x=7+2s \\ y=2+s \\ z=-6-3s \end{cases}$

② $-7 = 5t-s$
③ $-19 = 5t+3s$ −
$12 = -4s$
$s = -3$

Check $s=-3, t=-2$ in ①:

LS $= 4$

RS $= t-2s$
$= -2-2(-3)$
$= 4$

LS = RS

$\ell_2: \begin{cases} x=3+t \\ y=9+5t \\ z=13+5t \end{cases}$

$-7 = 5t-(-3)$
$-10 = 5t$
$t = -2$

POI: $x = 3+(-2) = 1$
$y = 9+5(-2) = -1$
$z = 13+5(-2) = 3$

POI is $(1,-1,3)$

d) $[x,y,z] = [3,1,2] + t[1,-4,-8]$
$4x+2y-z-8=0$

$l: \begin{cases} x=3+t \\ y=1-4t \\ z=2-8t \end{cases}$

$4(3+t)+2(1-4t)-(2-8t)-8=0$
$12+4t+2-8t-2+8t-8=0$
$4t=-4$
$t=-1$

POI:
$x = 3+(-1) = 2$
$y = 1-4(-1) = 5$
$z = 2-8(-1) = 10$

POI is $(2,5,10)$

e) $\ell: \begin{cases} x=2+t \\ y=2+2t \\ z=9+8t \end{cases}$

$\pi: 2x-5y+z-6=0$

$2(2+t)-5(2+2t)+(9+8t)-6=0$
$4+2t-10-10t+9+8t-6=0$
$0t=3$

No solutions. The line and plane are parallel and distinct.

f) $\ell: [x, y, z] = [3, -2, 1] + t[14, -5, -3]$
$\pi: x + y + 3z - 4 = 0$

$\ell: \begin{cases} x = 3 + 14t \\ y = -2 - 5t \\ z = 1 - 3t \end{cases}$

$3 + 14t + (-2 - 5t) + 3(1 - 3t) - 4 = 0$

$3 + 14t - 2 - 5t + 3 - 9t - 4 = 0$

$0t = 0$

Infinite solutions. The line and plane are parallel and coincident.

g) $\pi_1: x + 3y - 4z = 10$
$\pi_2: 3x + 9y - 12z = -30$

$3 \times$ ① $3x + 9y - 12z = 30$

② $3x + 9y - 12z = -30$ −

$0 = 60$

No solutions. The planes are parallel and distinct.

h) $\pi_1: 5x - 4y + z = 3$
$\pi_2: x + 3y = 9$

① $5x - 4y + z = 3$

$5 \times$ ② $5x + 15y = 45$ −

$-19y + z = -42$

Let $y = t$

$-19t + z = -42$

$z = -42 + 19t$

$x + 3t = 9$

$x = 9 - 3t$

$\ell: \begin{cases} x = 9 - 3t \\ y = t \\ z = -42 + 19t \end{cases}$

Line of intersection:

$[x, y, z] = [9, 0, -42] + t[-3, 1, 19]$

i) $\pi_1: x - 5y + 2z - 10 = 0$
$\pi_2: x + 7y - 2z + 6 = 0$
$\pi_3: 8x + 5y + z - 20 = 0$

① $x - 5y + 2z - 10 = 0$
② $x + 7y - 2z + 6 = 0$ −
④ $-12y + 4z - 16 = 0$

8 × ② $8x + 56y - 16z + 48 = 0$
③ $8x + 5y + z - 20 = 0$ −
⑤ $51y - 17z + 68 = 0$

17 × ④ $-204y + 68z = 272$
4 × ⑤ $204y - 68z = -272$
$0 = 0$

Line of intersection:

Let $y = t$

$-12t + 4z - 16 = 0$
$4z = 16 + 12t$
$z = 4 + 3t$

$x - 5y + 2z - 10 = 0$
$x - 5t + 2(4 + 3t) - 10 = 0$
$x - 5t + 8 + 6t - 10 = 0$
$x = 2 - t$

$[x, y, z] = [2, 0, 4] + t[-1, 1, 3]$

j) $\pi_1: 2x + y + 6z - 7 = 0$
$\pi_2: 3x + 4y + 3z + 8 = 0$
$\pi_3: x - 2y - 4z - 9 = 0$

① $2x + y + 6z - 7 = 0$
2 × ③ $2x - 4y - 8z - 18 = 0$ −
④ $5y + 14z + 11 = 0$

② $3x + 4y + 3z + 8 = 0$
3 × ③ $3x - 6y - 12z - 27 = 0$ −
⑤ $10y + 15z + 35 = 0$

2 × ④ $10y + 28z + 22 = 0$
⑤ $10y + 15z + 35 = 0$ −
$13z - 13 = 0$
$z = 1$
$5y + 14(1) + 11 = 0$
$y = -5$

$2x + y + 6z - 7 = 0$
$2x - 5 + 6(1) - 7 = 0$
$2x = 6$
$x = 3$

The POI is $(3, -5, 1)$

k) $\pi_1: x - y + 2z = 0$
$\pi_2: x - y + 2z + 7 = 0$
$\pi_3: z - 5 = 0$

No solution. Inconsistent system.
Planes 1 and 2 are parallel and distinct.

① $x - y + 2z = 0$
② $x - y + 2z = -7$ −
$0 = 7$

l) $\pi_1: x + 3y - z = -10$
$\pi_2: 2x + y + z = 8$
$\pi_3: x - 2y + 2z = -4$

$2\times$① $2x+6y-2z=-20$
② $2x+y+z=8$ −
④ $5y-3z=-28$

① $x+3y-z=-10$
③ $x-2y+2z=-4$ −
⑤ $5y-3z=-6$

④ $5y-3z=-28$
⑤ $5y-3z=-6$ −
$0=-22$

No solutions. Inconsistent system.
The planes intersect in pairs.

22) Find the distance between the point $P(3,-2,0)$ and the plane $4x - y + 8z = 2$.

Point on plane:
$4(0)-y+8(0)=2$
$y=-2$
$Q(0,-2,0)$
$\vec{QP}=[3,0,0]$
$\vec{n}=[4,-1,8]$

$$\text{distance} = |\text{proj}_{\vec{n}}\vec{QP}| = \frac{|\vec{QP}\cdot\vec{n}|}{|\vec{n}|} = \frac{12}{9} = \frac{4}{3} \simeq 1.33 \text{ units.}$$

23) Find the distance between these skewed lines.

$\ell_1: [x,y,z] = [1,0,-1] + s[2,3,-4]$

$\ell_2: [x,y,z] = [8,1,3] + t[4,-5,1]$

$P_1(1,0,-1)$
$P_2(8,1,3)$
$\vec{P_1P_2}=[7,1,4]$

$\vec{n}=\vec{m}_1\times\vec{m}_2=[-17,-18,-22]$

$3 \quad -5$
$-4 \quad 1$
$2 \quad 4$
$3 \quad -5$

$$\text{distance} = |\text{proj}_{\vec{n}}\vec{P_1P_2}| = \frac{|\vec{P_1P_2}\cdot\vec{n}|}{|\vec{n}|} = \frac{225}{\sqrt{1097}}$$

$\simeq 6.79$ units

1) $[x, y] = [2,7] + t[3,1]$
2)a) $[x, y] = [4,10] + t[1,1]$ **b)** $[x, y] = [-2, -8] + t[1, -13]$
3) Yes
4)a) $\ell: \begin{cases} x = 10 + 13t \\ y = 6 + t \end{cases}$ **b)** $\ell: \begin{cases} x = 3 + 6t \\ y = -9t \\ z = -1 + t \end{cases}$

5)a) $[x, y, z] = [1, -6,2] + t[4,1, -2]$ **b)** $[x, y, z] = [7,0,0] + t[0, -1,0]$

6) $[x, y, z] = [3,4, -5] + t[1, -1,2]$; $\ell: \begin{cases} x = 3 + t \\ y = 4 - t \\ z = -5 + 2t \end{cases}$

7) Parallel
8)a) $[x, y] = [-2,4] + t[-4,3]$ **b)** $[x, y, z] = [1,5,10] + t[0,0,1]$ **c)** $[x, y, z] = [-10,0,0] + t[3, -5, -9]$
d) $[x, y, z] = [3,0,0] + t[3,0, -1]$
9) Yes
10)a) x-int: $-\frac{7}{2}$; y-int: -7; z-int: 2 **b)** x-int: -6; y-int: none; z-int: -3

11) $\pi: \begin{cases} x = s \\ y = -4 + 10s + 3t \\ z = 1 - s + 4t \end{cases}$

12) $[x, y, z] = [2,0,11] + s[1,12,6] + t[7, -8,0]$

13) No
14) x-int: 1; y-int: -4; z-int: 3
15)a) $[x, y, z] = [1,3, -2] + s[2, -12,9] + t[3, -7,7]$; $\pi: \begin{cases} x = 1 + 2s + 3t \\ y = 3 - 12s - 7t \\ z = -2 + 9s + 7t \end{cases}$

b) $[x, y, z] = [8,0,0] + s[8,3,0] + t[4,0, -1]$; $\pi: \begin{cases} x = 8 + 8s + 4t \\ y = 3s \\ z = -t \end{cases}$

16)a) $y = 1$ **b)** $[x, y, z] = [2,1,1,] + s[3, -4,1] + t[-2, -2,3]$
17) $3x + 7y + z + 29 = 0$
18)a) $x - y - z - 1 = 0$ **b)** $5x + 6y + 8z - 61 = 0$
19)a) $[x, y, z] = [4, -2,7] + s[1,0,1] + t[0,3,5]$; $3x + 5y - 3z + 19 = 0$
b) $[x, y, z] = [-1, -2,5] + s[0,0,1] + t[0,1,0]$; $x = -1$ or $x + 1 = 0$
c) $[x, y, z] = [-3,7,1] + s[3,1,0] + t[1,0, -3]$; $3x - 9y + z + 71 = 0$
d) $[x, y, z] = [-2,3,12] + s[-2,1,5] + t[-3,7,8]$; $-27x + y - 11z + 75 = 0$
20) $96.9°$
21)a) $(356, -237)$ **b)** no intersection; skew lines **c)** $(1, -1,3)$ d) $(2,5,10)$
e) no intersection ; parallel and distinct **f)** infinite solutions; parallel and coincident
g) no intersection; parallel and distinct **h)** $[x, y, z] = [9,0, -42] + t[-3,1,19]$
i) $[x, y, z] = [0,2,10] + t[1, -1, -3]$ **j)** $(3, -5,1)$ **k)** inconsistent system; no solution
l) inconsistent system; no solution
22) $\frac{4}{3}$
23) 6.79